Wolverine SIH

Technical Report and Research Monograph

Abstract

The Wolverine ecosystem, developed for the Smart India Hackathon (SIH), is a deterministic prototype platform designed to demonstrate cross-platform intelligence gathering and entity resolution. Addressing the inherent isolation of distinct moderation regimes on heterogeneous web platforms, the project implements a simulated ecosystem of five distinct community networks (Atlas Market, Briar Bazaar, Cinder Exchange, Drift Forum, and Ember Commons) initialized with a synthetically generated dataset of 50,000 canonical personas.

This technical report documents the architectural, data, and analytical models implemented in the Wolverine prototype. Specifically, it details the pipeline for extracting, normalizing, and resolving identities across these disparate platforms using heuristic entity resolutionincluding Jaro-Winkler alias similarity and weighted temporal/behavioral cues. To mitigate data overload and assist human analysts, the system integrates a deterministic Retrieval-Augmented Generation (RAG) subsystem with strict prompt sanitization rules.

Crucially, this monograph explicitly delineates the boundaries of the prototype. While the system architecture demonstrates advanced conceptssuch as multi-hop graph traversal and cryptographic trust layerskey elements like the graph projection and evaluation metrics are implemented as in-memory simulations or mocked outputs designed strictly for controlled demonstration. By strictly separating verified implementation from conceptual specifications, this report provides a rigorously honest academic evaluation of the Wolverine SIH submission.

Acknowledgements

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List of Abbreviations

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Chapter 1

Introduction

The fragmentation of online communities has inadvertently created isolated silos of user behavior, identity, and moderation. When individuals interact across heterogeneous platformsranging from clear web forums to decentralized marketplacestheir digital footprints are fractured. For intelligence analysts, trust and safety teams, and law enforcement, this poses a critical challenge: bad actors can exploit the boundary lines between distinct moderation regimes to obfuscate their activities, knowing that reputation and threat indicators rarely cross platform borders.

1.1 Problem Context

The core problem addressed by the Wolverine project is the inability to securely and accurately trace threat actors across structurally independent platforms. In real-world environments, each platform enforces its own data schema, access controls, and moderation policies. This isolation means that an entity exhibiting malicious behavior on one forum may simultaneously maintain a pristine reputation on an adjacent marketplace. The lack of cross-platform visibility severely hinders the proactive identification of coordinated campaigns, illicit supply chains, and distributed threat networks.

1.2 Motivation

Bridging this visibility gap requires a mechanism to correlate identities across boundaries without necessarily relying on explicit personally identifiable information (PII). However, building such a system in a production environment introduces immense technical and ethical complexities, ranging from massive data normalization bottlenecks to privacy and data sovereignty concerns. The motivation for the Wolverine Smart India Hackathon (SIH) prototype is to create a deterministic, bounded environment to model and demonstrate how cross-platform identity resolution could operate heuristically on disparate intelligence streams.

1.3 Objectives

The Wolverine project was designed with the following technical objectives:

- **Synthetic Ecosystem Construction:** Generate and operate a controlled, closed-loop ecosystem of five heterogeneous platforms (using diverse technology stacks) to serve as intelligence sources.

- **Data Normalization Pipeline:** Ingest raw, semi-structured JSON payloads from these isolated platforms and map them to a canonical intelligence schema.

- **Heuristic Entity Resolution:** Implement algorithmic similarity scoring (such as Jaro-Winkler string distance) to probabilistically link user aliases and digital artifacts across the platforms.
- **Analyst Augmentation:** Provide a unified, read-only analyst interface integrated with a sanitized Retrieval-Augmented Generation (RAG) subsystem to summarize findings deterministically.

1.4 Scope and Prototype Constraints

It is critical to explicitly define the boundaries of the Wolverine prototype. The system was engineered specifically as an interactive demonstration for the SIH evaluation rather than a production-ready intelligence platform.

Consequently, several operational constraints apply:

- **Synthetic Data:** The entire ecosystem operates on a synthetic dataset of 50,000 personas; no real-world, live OSINT (Open Source Intelligence) data is ingested.
- **In-Memory Processing:** While the architecture theorizes advanced PostgreSQL recursive CTEs for graph traversal, the verified implementation relies on an in-memory Breadth-First Search (BFS) bounded by a single host's memory.
- **Simulated Evaluation:** Evaluation metrics (e.g., F1 scores and precision) presented in the project documentation are static, hardcoded outputs from a mock evaluation harness rather than dynamically calculated performance measures.

These limitations are deliberately documented to ensure academic honesty and to separate the conceptual architecture from the implemented demonstration reality.

1.5 Contributions

Despite its prototype constraints, the Wolverine project makes the following demonstrative contributions:

1. A fully executable, 14-container Docker ecosystem simulating a fragmented dark web/clear web environment.
2. An 11-phase deterministic data generation pipeline capable of constructing coherent, cross-platform behavioral footprints.
3. A functional heuristic entity resolution pipeline that normalizes arbitrary social data into a canonical graph format.

1.6 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 establishes the domain and technical foundations of cross-platform intelligence. **??** reviews related work in entity resolution and synthetic data generation. **??** outlines the system requirements and threat models governing the prototype. **??** and **??** detail the overarching system topology and the Wolverine Database design, respectively. **??** dissects the implemented heuristic algorithms and LLM integrations. **????** describe the concrete implementation details and the synthetic demonstration environment. **????** analyze the evaluation framework and ethical constraints. Finally, **??????** provide a critical discussion, outline future directions for scaling, and conclude the report.

Chapter 2

Domain and Technical Foundations

To contextualize the Wolverine prototype, this chapter outlines the foundational concepts in cross-platform identity intelligence, entity resolution, and the specific data structures utilized within the system's simulated ecosystem.

2.1 Relevant Domain Concepts

Modern intelligence gathering increasingly relies on analyzing digital artifacts scattered across independent domains. In the context of the Wolverine prototype, the primary domain is modeled after Open Source Intelligence (OSINT) and dark web monitoring. Key concepts include:

- **Heterogeneous Silos:** Independent digital communities (forums, marketplaces, social networks) that enforce distinct schemas and share no underlying infrastructure.
- **Digital Footprints:** The trail of metadata left by a user, including aliases, timestamps, IP addresses (or proxies), and interaction patterns.
- **Cross-Platform Correlation:** The analytical process of deducing that two distinct digital personas on different platforms belong to the same real-world entity.

2.2 Data and Entity Representation

A significant challenge in correlating heterogeneous data is structural incompatibility. Platform A might represent a user interaction as a nested JSON object involving `buyer_id` and `seller_id`, whereas Platform B uses a flat relational schema of `author` and `post`.

Wolverine addresses this by enforcing a *Canonical Record Schema* during the ingestion pipeline. Raw captures are mapped into a standardized format containing:

- **Entity Nodes:** Representations of personas, digital artifacts (e.g., cryptocurrency wallets), or content.
- **Relational Edges:** Defined interactions (e.g., `AUTHORED`, `INTERACTED_WITH`, `TRANSACTED`).
- **Temporal Metadata:** Standardized ISO-8601 timestamps used to evaluate proximity constraints.

This canonical representation is the foundational layer upon which the heuristic resolution algorithms operate.

2.3 Graph Concepts in Wolverine

The canonical data model naturally forms a directed graph where personas and artifacts are nodes, and their activities form the connecting edges. In theoretical production deployments, resolving identities across such a network requires highly scalable graph traversal methodologies, often leveraging specialized graph databases or recursive Common Table Expressions (CTEs) in relational systems like PostgreSQL.

However, as a prototype, Wolverine implements a computationally bounded approach:

- **In-Memory Breadth-First Search (BFS):** Graph projection and neighborhood traversal are executed entirely in the memory space of the Node.js application process.
- **Bounded Depth:** Traversals are strictly limited to a shallow depth (typically 2 to 4 hops) to prevent memory exhaustion and infinite loops in cyclic interaction patterns.

This foundation enables the demonstration of multi-hop intelligence correlation while deliberately constraining the operational footprint to a single host.

2.4 Relevant Computational Methods

To determine if two distinct aliases might represent the same entity, Wolverine employs deterministic heuristic similarity scoring. Rather than relying on computationally expensive or unpredictable Machine Learning (ML) models, the prototype uses the following formal methods:

2.4.1 Jaro-Winkler Distance

The Jaro-Winkler string metric is a measure of edit distance between two sequences. It is particularly effective for short strings such as usernames or aliases, as it heavily weights prefix matches. The Jaro-Winkler similarity s_{jw} is bounded between 0 and 1, where 1 equates to an exact match. Wolverine uses this metric as the primary driver for computing ‘aliasSimilarity‘.

2.4.2 Temporal Proximity Decay

Behavioral correlation assumes that coordinated actions across platforms often occur within narrow time windows. Wolverine applies a linear decay function to calculate a temporal proximity score based on the day difference between account creations or specific activities. As the temporal distance increases, the proximity score linearly decays toward zero.

2.4.3 Heuristic Weighting

The final resolution score is a weighted linear combination of distinct similarity features (alias, display name, temporal proximity, and behavioral overlap). By applying static, predefined weights to these normalized features, the system computes a deterministic confidence score that governs whether two nodes should be linked in the graph. *Detailed formulations of these models, including prototype-specific static constants, are provided in* **??**.

Chapter 3

Related Work and Research Context

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System Requirements and Threat Model

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Synthetic Generator Configurations and Schema

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Appendix B

API Specifications and Protocol Adapters

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Appendix C

Wolverine DB Formal Specifications (Abbreviated)

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Appendix D

Evaluation Harness Mock Outputs

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Bibliography